

Amitabh

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Team Lead and RF System architect with 10+ years of experience in designing and developing advanced RF solutions. Experienced in designing, testing, and validating RF receiver and transmitter boards. Responsible for the design, implementation, schematic capture, PCB guidance, electrical circuit design, testing, optimization, and debugging of circuits, including driver amplifiers, GaN amplifiers, and transmitter and receiver chain front-end blocks and filters.

Possesses excellent interpersonal and communication skills alongside a broad range of technical abilities. Enjoys being part of a team, as well as managing, motivating, and training a productive unit, and thrives in a high-pressure, challenging work environment.

Technical Skills

RF hardware board design, LNA design for 5G applications, knowledge of Ethernet and digital circuit design, power supply design for DC, independent component testing, system integration testing, subsystem and integrated testing.

RF Equipment

Spectrum Analyzer, Network Analyzer, Signal Generator, Arbitrary Waveform Generator, Power Meter, Analog, Digital & Mixed Oscilloscope, Impedance Analyzer, Power Meter, Multimeter, and LCR meter.

Software/ Tools

- Excel-Advanced, Linx Environment, Python, Matlab-Simulink, C Programming, Keysight-ADS, Cadence-AWR, HFSS, OrCAD, Allegro PCB Design, and ADIsimRF, Arbitrary wave generator and Transceiver **Proteus Arbitrary Waveform Transceiver (AWT)**

Work Experience

Company	Designation	Duration	Achievement
Dhruva Space	Sr. Manager (Tech Lead)	Nov 2025-Present	
Tabor Electronics	FAE-Global	June 2025-Nov2025	Technical Document Proofreading
Skyroot Aerospace	RF Lead	July 2021-Sept 2024	RF lead for India's first Private launch vehicle
Bharat Forge	RF Lead	Sept 2017-June 2021	Established an RF and Radar lab in Kaylani Group
IIT Bombay	RF Research Engineer	Sep 2016- March 2017	R&D
IIT Madras	RF Research Engineer	Sept2015-August 2016	R&D

Qualification

Indian Institute of Information Technology Design and Manufacturing (IIITD&M), Kancheepuram, Chennai Master of Design (M-Des), Electronic System Design. 2013-2015

Project Summary

Dhruva Space: Nov 2025-Present

Designation: Sr. Manager (Tech Lead)

- X-band Link budget for a high data rate transmitter
- TTC link budget
- Leading In-house High Gain X-band and S-band antenna for the SATCOM satellite
- Leading In-House development of deployable helical antenna for IoT applications
- EGS Test plan and documentation
- Working on ground and space-based IoT network
- Antenna selection for IoT nodes and terrestrial nodes

Tabor Electronics: June 2025- Nov 2025

Designation: Product FAE-Global

Proteus Arbitrary Waveform Transceiver (AWT)

- Hands-on experience with **Proteus Arbitrary Waveform Transceiver (AWT)**
- **Python/MATLAB API** development for AWT-based solutions
- Experience in **SCPI** programming and **Instrument Automation**
- **IQ-based** signal modulation and demodulation

Skyroot Aerospace: July 2021- Sept 2024

Designation: RF Lead

Antenna development for launch vehicle

- **L-Band RHCP** antenna design and development for GPS application
- C-Band **Inverted F -antenna** development for Transponder application
- S-Band Inverted F antenna development for telemetry application
- UHF -Band Inverted F- antenna development for Telecommand
- **Thermal analysis and thermal qualification** for antenna
- **Antenna test and Qualification**
- Directed the finalization of RF architecture for ground stations.
- Conducting comprehensive RF link budget analyses and **coverage factors** to guarantee optimal performance.
- RF Shield and Mechanical Enclosure Analysis for **low weight and cost optimization**

Low-noise Amplifier development for a GPS receiver

- Finalisation of LNA specification and component
- Circuit Design, Schematic development, and Circuit Simulation **using PDK file for non-linearities on ADS**
- Designed a **Broadband matching circuit**
- **Board layout** planning and development, **EM and EM -co simulation** using **RF-Pro Tool** for signal integrity
- Performance measurements by measuring **S-parameter, P1, IP3, SFDR, SINAD and Noise figure**.

Power Amplifier development for S-band telemetry

- Finalisation of **GAN-based 15-watt power Amplifier** specification and components
- Circuit Design, Schematic development, and Circuit Simulation **using PDK file for non-linearities on ADS**
- **Basing circuit design and Load pull simulation**
- **Board layout** planning and development, **EM and EM -co simulation** using **RF-Pro Tool** for signal integrity
- Performance measurements by measuring **low signal gain and saturated power**

Bharat Forge: Sept 2017- July 2021

Designation: RF lead

Development of a Private 5G Network for Battlefield Applications

- Designed and optimized 5G NR RF subsystems **for a private 5G network, achieving a 5 km range in sub-6 GHz (3.5 GHz)**
- **Led RF performance testing** for transmitter, receiver, and antenna subsystems, validating **EVM, ACLR, and sensitivity.**
- Performed end-to-end **link budget** analysis
- Developed RF test cases and measurement setup as per **3GPP standard**
- Doherty Power Amplifier: Simulated advanced **Doherty power** amplifiers and associated components
- **CCFD** measurement, **PAPR** assessment, **IQ impairment**

Development of UWB radar

- Designed and developed a **sub-nanosecond short pulse** ultra-wideband (UWB) **transmitter** for radar applications.
- Designed and developed a **dual-gate RF MOSFET mixer.**
- Designed and developed an **RF frontend** tailored for **Ku-band** applications.
- **System Integration** and **EMI** Management, and Testing
- MIL Standard-461

IIT Bombay and IIT Madras

- Designed and developed a **coupled line filter**
- Designed a **Hybrid coupler**
- Developed a **zero-index metamaterial** for antenna gain enhancement
- Designed and developed a **hairpin filter**